

LOs	Grade				
	NA	Resub	P	M	D
LO1	☐	☐	☐	☐	☐
LO2	☐	☐	☐	☐	☐
LO3	☐	☐	☐	☐	☐

Student Name				Code								
Student Section					☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐							
Unit No. & Title	ICT - D - 425 Machine Learning				☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐							
Qualification	Bachelor of Technology in Information Technology (S/W development Track) (y4-2nd semester)				☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐							
Assignment Part / No.	1&2				☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐							
Hand out date	4/2026	Hand in date	5/2026		☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐							
Assessor Name	Dr.Rasha	IV Name	Prof.assistant/Eman mounir		☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐							

LO No.	Criteria	Task No.	Grade	Comments	LO No.	Criteria	Task No.	Grade	Comments
Lo1	P	P1	1	☐	Lo2	P	P4	2	☐
		P2	1	☐			P5	2	☐
		P3	1	☐			P6	2	☐
	M	M1	1	☐		M	M3	2	☐
		M2	1	☐			D2	2	☐
	D	D1	1	☐		D	D2	2	☐
Lo3	P	P7	3	☐					
	M	M4	3	☐					
	D	D3	3	☐					

"I certify that this assignment is my own work, written in my own words. Any other person's work included in my assignment is referenced / acknowledged".

IV Signature

Eman Monir

Rasha

Date: 4/2026

LO No.	Targeted criteria	To achieve the criteria the evidence must show that the student is able to:	Evidence
LO1	P	P1 Recognize how machine learning (ML), deep learning, and generative artificial intelligence (generative AI) are part of artificial intelligence (AI)	DOCUMENT
		P2 Describe artificial intelligence and machine learning terminology&challenges	
		P3 Identify how machine learning can be used to solve a business problem	
	M	M1 Describe the machine learning process	
		M2 List the tools available to data scientists.	
	D	D1 Identify when to use machine learning instead of traditional software development methods	
LO2	P	P4 Formulate a problem from a business request	DOCUMENT
		P5 Build a Jupyter Notebook using Jupyter or any other popular Framework	
		P6 Outline the process for evaluating& visualize data	
	M	M3 Use open source tools to examine and preprocess&Feature engineering selected dataset	
	D	D2 Use Amazon or Colab tool to train a ML model	
LO3	P	P7 Use ROC, Sensitivity,Cross-validation methods to test the performance of ML model	DOCUMENT
	M	M4 Use a hosted model for inference	
	D	D3 Create hyperparameter tuning job to optimize a model's effectiveness	

Scenario

You are AI analyst, for a classification, regression, clustering problems such as:

Boston Housing Dataset

Student Performance Dataset

Heart Disease DatasetPima

Indians Diabetes Dataset

Iris Flower Dataset

Glass Identification Dataset

Spam Email Dataset

Breast Cancer Dataset, using sources from Kaggle,UCI Machine Learning Repository,Google Dataset Search

your supervisor asked you to answer the following tasks:

Task No.01

1 Recognize how machine learning (ML), deep learning, and generative artificial intelligence (generative AI) are part of artificial intelligence (AI) p1

2 Describe artificial intelligence and machine learning terminology&challenges p2

3 Identify how machine learning can be used to solve a business problem p3

4 Describe the machine learning process m1

5 List the tools available to data scientists.m2

6 Identify when to use machine learning instead of traditional software development methods d1

Task No.02

1 Formulate a problem from a business request p4

2 Build a Jupyter Notebook using Jupyter or any other popular Frameworkp5

3 Outline the process for evaluating& visualize data p6

4 Use open source tools to examine and preprocess&Feature engineering selected dataset m3

5 Use Amazon or Colab tool to train a ML model d2

Task No.03

1 Use ROC, Sensitivity,Cross-validation methods to test the performance of ML model p7

2 Use a hosted model for inference m4

3 Create hyperparameter tuning job to optimize a model's effectiveness d3

Project Deliverables (Evidence)

Students must submit **one PDF document** that includes all required project work.

The PDF must contain:

- **Project Introduction** (scenario and objectives)
- **Dataset**
- **Explanation of Concepts**



- Dataset preprocessing screenshots
 - Dataset training
 - Dataset Testing
 - Case Study deployment&Evaluation
-
- Students must **demonstrate on pcs connected with internet**
 - The system must be **fully working and executable**
 - Students should be ready to:
 - Run Code
 - Explain all project steps

Resubmission Feedback: *Please note resubmission feedback is focussed only on the resubmitted work	
Assessor Signature: <i>Rasha</i>	Date :
Internal Verifier's Comments:	
IV Signature: <i>Eman Monir</i>	Date :